

Thomas Xiao

Irvine, CA | thomasxi@andrew.cmu.edu | LinkedIn | GitHub | Portfolio

EDUCATION

Carnegie Mellon University

B.S. in Mathematical Sciences; Computational and Applied Mathematics

Arnold O. Beckman High School

High School Diploma; Weighted GPA: 4.63/5.00; SAT: 1570

Pittsburgh, PA

Aug. 2026 – May 2030

Irvine, CA

Aug. 2022 – May 2026

RESEARCH EXPERIENCE

Independent Climate Research Collaboration

Research Collaborator

Remote

2024 – 2025

- Analyzed NOAA U.S. climate records from 1895–2024 with Python, harmonic regression, Fourier analysis, wavelet decomposition, and change-point detection.
- Built reproducible data-cleaning, validation, modeling, and visualization workflows; compared frequency-aware models with trend-only baselines across climate variables and regions.

USC Marshall School of Business

Research Assistant – Game Design and Consumer Behavior

Remote

2024 – 2025

- Collaborated with Dr. Cosimo Arnesano to analyze BoardGameGeek data using regression and gradient-boosting methods.
- Evaluated relationships among ratings, complexity, play time, and consumer response; documented findings for product-design and marketing decisions and presented at CONF-APMM 2025.

SOFTWARE ENGINEERING PROJECTS

Personal Portfolio and Online Resume | *React, JavaScript, HTML/CSS, GitHub Pages*

2024 – Present

- Built and deployed a responsive React portfolio that organizes research, projects, technical skills, and resume links into reusable page components.
- Customized an open-source GitProfile experience with project cards, research pages, theme switching, responsive layouts, and direct links to code, papers, and demos.

Desmos Bezier Renderer | *Python, Flask, OpenCV, NumPy, Potrace*

2023 – Present

- Extended a local image/video processing workflow that applies Canny edge detection and Potrace tracing to convert frames into Bezier-curve expressions for Desmos.
- Implemented a Flask backend with configurable frame processing, filtering, output color, image export, and a browser-based calculator frontend.

M3 Challenge Model | *Python, NumPy, pandas, Matplotlib, LaTeX*

2025

- Built a mathematical model for urban heat, power demand, and community vulnerability in the 2025 MathWorks Math Modeling Challenge.
- Translated assumptions and data into a reproducible analysis and technical paper with Python visualizations and LaTeX documentation.

LEADERSHIP AND EXPERIENCE

Beckman Computer Science Club

Board and Program Lead

Irvine, CA

2024 – 2026

- Taught algorithms and data structures to 15+ students and mentored peers building Python and Discord applications.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, MATLAB, SQL **Frameworks/Tools:** React, Flask, Streamlit, Git, GitHub Pages, LaTeX, Jupyter

Libraries: NumPy, pandas, Matplotlib, Seaborn, OpenCV, scikit-learn, PyWavelets **Methods:** Regression, gradient boosting, Fourier analysis, wavelets, change-point detection, data visualization

Honors: HiMCM Honorable Mention (top 28% of 1,055 teams); 2nd Place, USC SCMC Math Competition and CSULB Math Day at the Beach